

Doppel Market

Market Research Report

MS Group 1:

Sierra Bao, Zixin Meng, Sarah Cho, Xiuhan Chen, Leon Tang



Executive Summary

| Qualitative Insights

- Online use is **frequent and low-effort**, centered on scrolling, streaming, and shopping—any solution must fit existing routines and clearly save time or money.
- Participants are aware their data is tracked but **do not clearly understand by whom or for what purposes**; they expect plain-language explanations of data use and tangible benefits in return.
- Participants are more comfortable sharing **lower-risk data** than sensitive information and expect tiered consent, low-effort participation, and strong safeguards to trust a digital twin or research platform.

| Quantitative Insights

- **Adoption potential is strong**, but hesitation is high: 38.1% likely, 33.3% unlikely, 28.6% undecided.
- **Financial incentives drive research participation**, but **are difficult to drive digital-twin adoption**.
- **Adoption barriers are trust-based, not solely financial**—users fear misuse, loss of control, and data breaches.
- **Clear privacy guarantees and transparent data governance** are essential to convert undecided users.

Executive Summary: Strategic Recommendations

“ *Building Trust* ”

- Add clear “**trust moments**” and simplifying privacy explanations with short, visual cues.
- **Reinforce transparency** by showing users monthly summaries and **visible outcomes** from their participation.
- Strengthen legitimacy by partnering with **trusted community organizations** such as libraries and unions.

“ *Building Value* ”

- Highlight the digital twin as a **practical time-saving shortcut** and show estimated hours saved in the UI.
- Motivate participation by **offering small, fast, guaranteed rewards** with flexible redemption options.
- Drive adoption through community-based **referrals** using simple sharing tools and small referral credits.

Project Objectives

Explore whether digitally engaged individuals from lower socioeconomic backgrounds, who are more financially motivated to participate in data-sharing platforms, represent a viable early-adopter segment for Doppel Market's digital twin platform.

In this project, lower socioeconomic backgrounds refers to individuals who are positioned toward the lower end of the socioeconomic spectrum based on income, education, and occupation— for example, people in lower-wage, hourly, or frontline/service jobs (e.g., building staff, retail, delivery drivers) with relatively limited financial cushion compared to the broader local population.

Willingness to Participate

How willing this group is to create and maintain a digital twin, and what specific conditions (data control, transparency, effort level) are required for them to say yes and stay engaged.

Value Exchange & Incentives

What “fair” compensation looks like for different levels of data sharing, and whether financial rewards meaningfully shift willingness relative to concerns about privacy and misuse.

Privacy Boundaries & Trust Drivers

Which data types feel acceptable vs off-limits, how much control they expect over sharing and deletion, and what signals (language, safeguards, brand cues) build enough trust for Doppel to be seen as a credible, safe platform.

Methodology Overview

| Qualitative Phase

- Conducted in-person interviews using convenience sampling
- Each interview lasted 20-30 minutes and participants received a \$10 Amazon or Starbucks gift card as incentive.
- Used a shared core interview script to ensure consistency across participants, while keeping conversations flexible enough to probe (*“tell me more”* or *“how did that make you feel?”*), reorder questions, and follow emergent themes in respondents' own words.

The key questions included:

1. **Online Behavior:** “Thinking about a typical day online, what kinds of things do you usually do?”
2. **Data & Privacy:** “How do you feel about sharing your personal information with apps and websites?”
3. **Research Participation:** “Have you ever participated in research studies or surveys before? What made you say yes or no?”
4. **Digital Identity / Digital Twin:** “What comes to mind when you hear ‘digital twin’ or a ‘digital copy of yourself’?”

| Quantitative Phase

- Cross-sectional quantitative study using the Qualtrics survey with non-probability convenience sampling
- Screening questions ensured only target participants in low SES proceeded
- Participants were recruited directly at high-traffic residential and commercial touchpoints, including apartment concierges, mailrooms, and retail outlets.
- The instrument was a 28-item structured survey combining closed-ended items (5-point Likert-type scales, frequency scales, multi-select checklists, and ranking tasks) with a small number of brief open-ended questions.

The contents included:

1. **Online and AI usage patterns**, general privacy concern, and trust in data-handling entities.
2. **Past research participation**, motivations/barriers, and preferred incentives.
3. **Possible drivers to adopt a digital twin** (after providing an in-survey definition), perceived risks (e.g., liability/misuse), and comfort sharing different data types.

Respondent Profile

| Qualitative Phase

- The qualitative sample consists of 5 adults recruited via convenience sampling from our own buildings, gyms, and workplaces.
- All five respondents work in frontline/service roles (doorman, gym front desk/concierge, retail staff, building security); in this study, we use these occupations as a proxy for lower socioeconomic status.
- The sample is 60% female and 40% male, and all reported regular smartphone and app use (e.g., social media, streaming, online shopping).

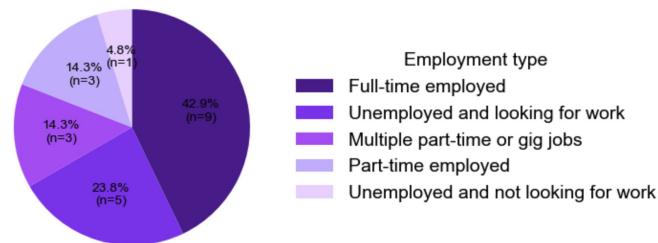
| Quantitative Phase

- The sample consists of 21 respondents, primarily working-age adults, with most falling between 18–44 years old.
- Employment varies, but 43% are full-time employed, and key industries include warehouse/transportation and administrative support (each 24%).
- The sample is 48% Asian, 24% White, and 19% Hispanic/Latino, with a balanced gender split (48% female, 43% male).
- Education levels are relatively high, with 38% holding a bachelor's degree and 33% having some college experience.

Table 1.1

Warehouse, delivery, transportation, or gig/app work	5	23.81
Administrative or clerical support	5	23.81
Retail, food service, or hospitality	2	9.52
Manufacturing, maintenance, or skilled trades	2	9.52
Home health, childcare, or personal care services	2	9.52
Security	1	4.76
Influencer	1	4.76
Research Assistant	1	4.76
Gym concierge	1	4.76
Other service job (please specify):	1	4.76

Table 1.2



Qualitative Research

Research Objectives

- Assess willingness to **create and maintain a digital twin for market research** and how respondents would prefer it to function relative to traditional surveys or panels.
- Explore expectations around **data access, control, and sharing**, including comfort with data being used by retailers and third parties.
- Gauge perceptions of **data monetization and fair value exchange**—what types of rewards or payments feel appropriate for different kinds and levels of data sharing.

Key Interview Questions

Online Behavior

- “Thinking about a typical day online, what kinds of things do you usually do?”
- “When you use these apps or platforms, what do you think is happening with your activity and data?”

Data & Privacy

- “How do you feel about sharing your personal information with apps and websites?”
- “Can you recall a recent moment when you noticed how your data was being used online? What came to mind?”

Research Participation

- “Have you ever participated in research studies or surveys before? What made you say yes or no?”
- “What would need to be different for you to consider participating (or participating more often)?”

Digital Identity / Digital Twin

- “What comes to mind when you hear ‘digital twin’ or a ‘digital copy of yourself’?”
- “If you were to create something like this, what would you want to know first, and what excites or concerns you about it?”

Qualitative Research

Key Insights

- **Digital life is high-frequency but low-effort.**

Online behavior is dominated by scrolling, streaming, and shopping; respondents mostly consume rather than create, so any solution must fit existing routines and clearly save time, money, or effort.

- **Data feels tracked, but not understood.**

Participants know their data is being collected but cannot clearly say by whom or for what; they mentally separate “lower-risk” data (demographics, purchases, entertainment) from sensitive data (finance, health, precise location) and expect plain-language explanations and tiered consent.

- **Research participation is a conditional trade, not a habit.**

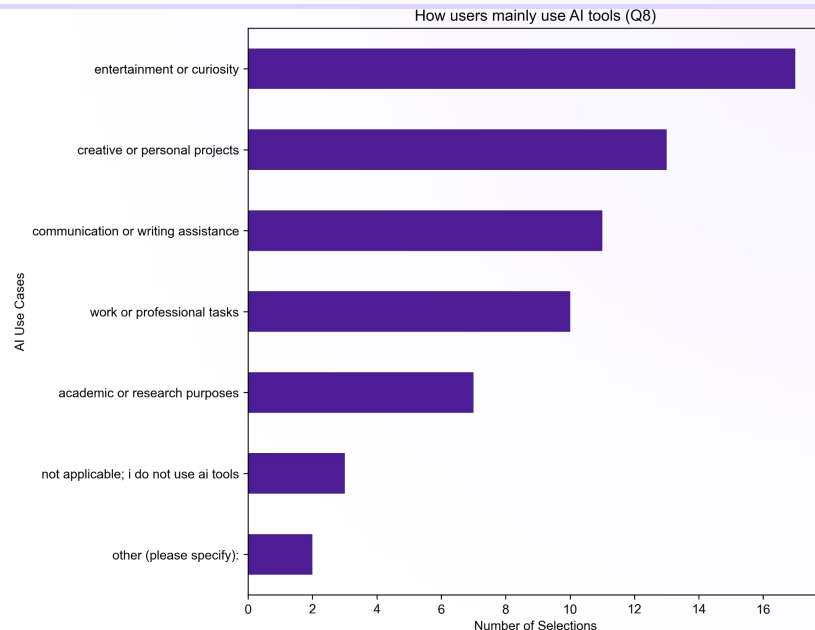
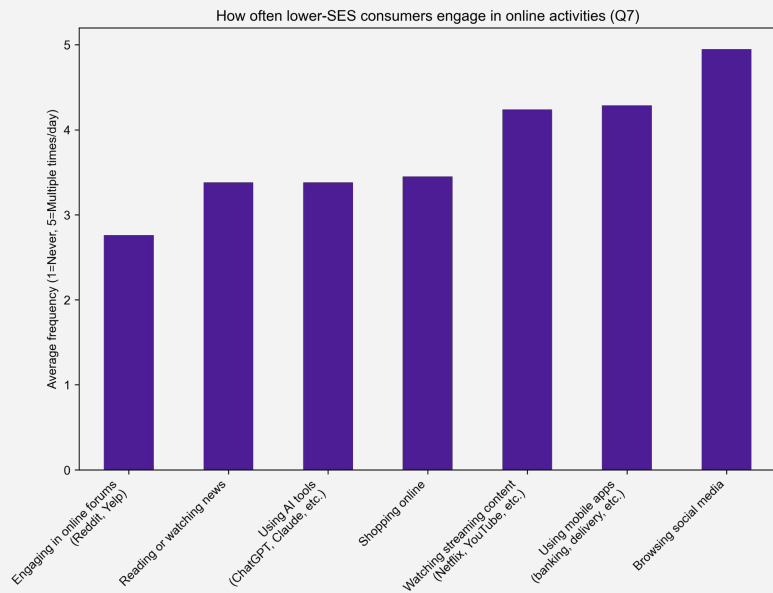
Taking part in studies is occasional and opportunistic, driven by convenience and immediate payoff; ongoing participation depends on low burden, clear personal relevance, and visible outcomes so they don’t feel like “test subjects.”

- **A digital twin is acceptable only as a controlled, helpful proxy.**

The term initially triggers confusion and risk– many respondents don’t actually know what “digital twin” means and associate it with cloning or video-game avatars– but interest grows when it is framed as a practical tool that can “answer like them” under strong safeguards, with trust built through transparency, explicit opt-ins, and a fair value exchange.

Quantitative Research: Online Presence Overview

- **Entertainment and convenience drive digital engagement:** social media, mobile apps, and video streaming dominate daily digital habits.
- **Knowledge-driven platforms are less embedded:** news consumption lag behind.
- **AI has not reached habitual use:** AI tools shows low frequency



- **AI is primarily used for entertainment, curiosity, and creative projects**, reflecting low-stakes, personally motivated engagement rather than productivity or problem-solving.
- **Communication and writing support are key use cases**, helping users reduce effort, improve clarity, and feel more confident in expression.
- **Work, academic, or professional applications remain secondary**, indicating that most users treat AI as a supportive assistant instead of a full productivity tool.

Quantitative Research: Online Presence vs. Privacy Concerns



Social media use shows the strongest positive privacy concern correlation. Heavy social media users are more likely to think about privacy and worry about how companies use their data (Q9 & Q11).

Online shopping users acknowledge data collection but are not alarmed by it. They understand cookies, recommendations, and personalization, but see them as transactional and low-risk.

AI tool users are privacy-aware, not privacy-fearful. They think about data usage (Q9), yet report little concern or anxiety toward data collection/use (Q10 & Q11).

News consumption shows the most negative correlations across all privacy dimensions.

Frequent news users appear privacy-fatigued or cynical — they disengage rather than worry about data practices.

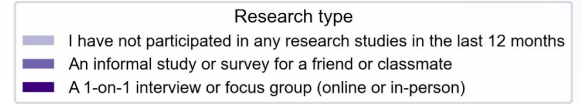
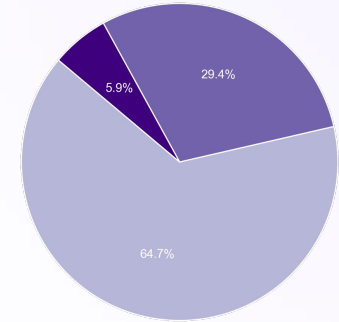
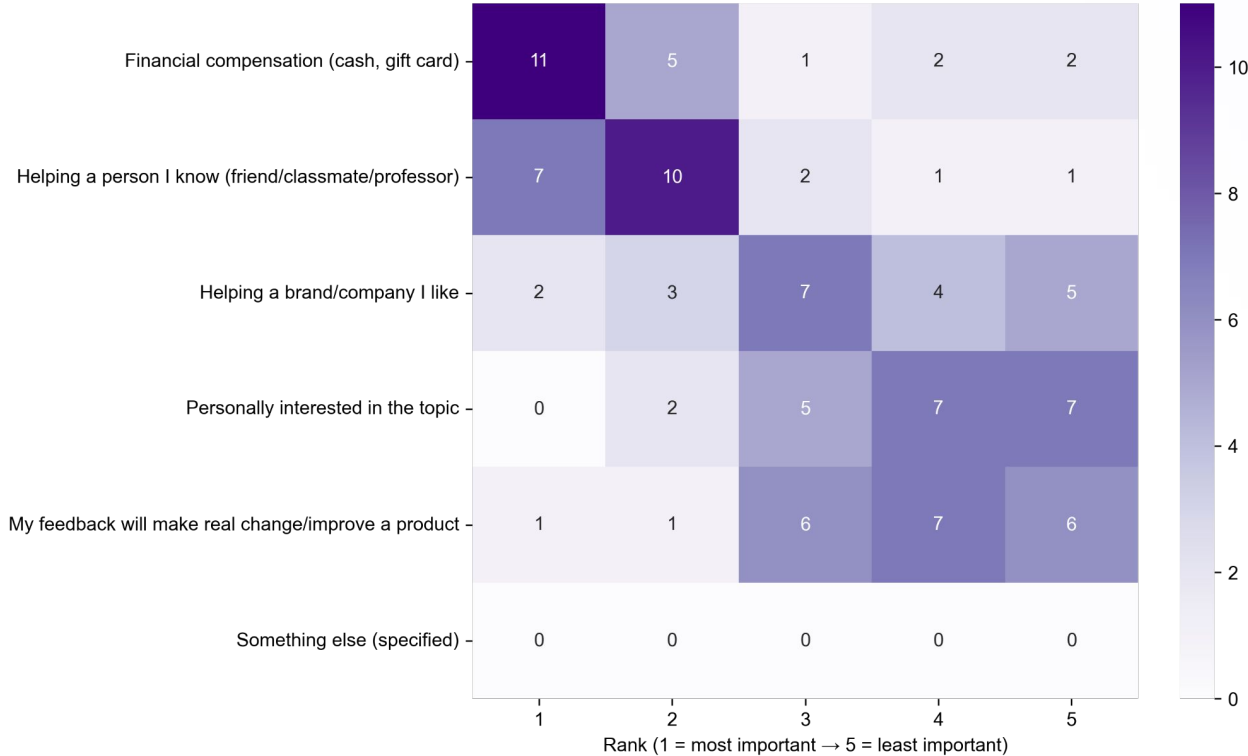
Strategic Takeaway:

Privacy behavior is emotional, not rational.

Users judge risk based on how their digital habits make them *feel*, not on how their data is actually processed.

Quantitative Research: Core Motivators Behind Survey Participation

Rank distribution of factors influencing participation (Q17)



Social capital matters

Helping someone you personally know is disproportionately motivating.

Money is universal

It bypasses all psychological filters and is effective across groups.

Intrinsic motivations are weak

Curiosity or interest in the research topic is not enough for most.

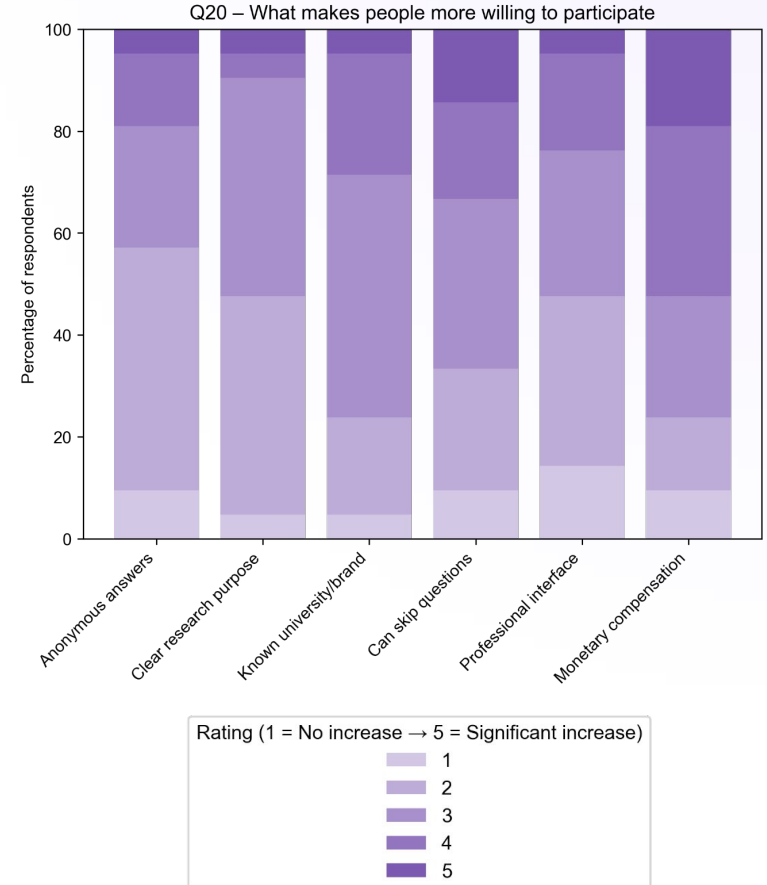
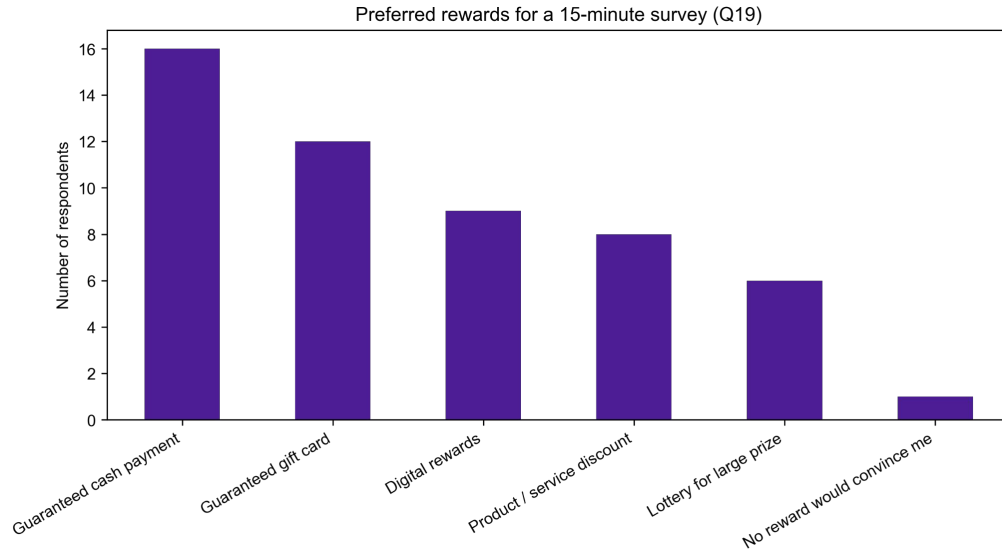
Quantitative Research: Core Motivators Behind Survey Participation

Guaranteed money is the most powerful motivator.

Respondents clearly prefer immediate, tangible rewards over abstract or delayed benefits.

Autonomy boosts willingness.

Allowing participants to skip uncomfortable or irrelevant questions makes them more likely to engage.



Quantitative Research:

User Willingness to Delegate Tasks to a Digital Twin During Research Participation

Highest Acceptance Features:

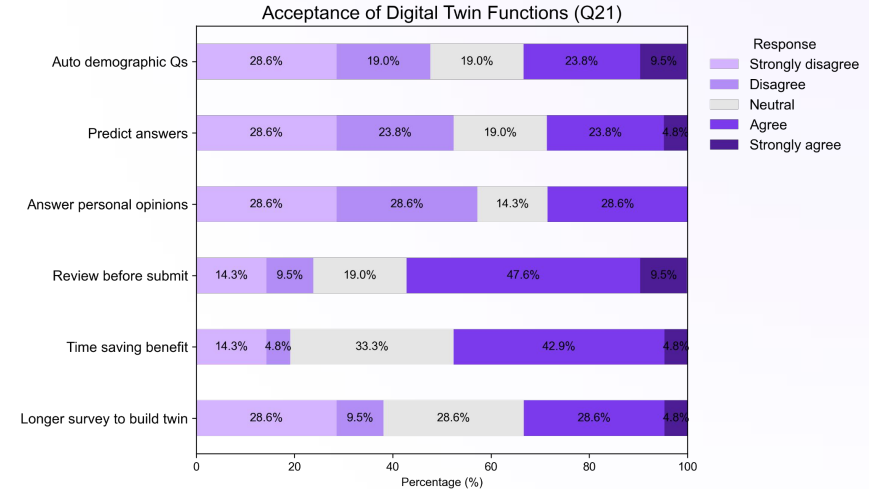
- **Time-saving benefit** shows the one of the strongest agreement: 47.7% agree/strongly agree that avoiding repetitive questions is valuable, with only 19.1 % disagreeing, which provides insights for strongest value proposition.
- **Review before submit** is the most important trust feature: 57.1% want approval control, suggesting users desire human oversight despite automation benefits.

Moderate Willingness:

- **Auto demographic Qs:** 33.3% would let digital twins handle basic demographics, while 47.6% remain uncomfortable. This indicates cautious openness to low-risk automation.
- **Predict Answers:** Split between willing (28.6%) and unwilling (52.4%), with 19% neutral. This suggests skepticism about prediction accuracy remains a barrier.

Strategic Takeaway:

When designing function, start with low-risk, time-saving automation and build trust through human-in-the-loop controls; avoid asking digital twins to express personal opinions until strong trust is established.

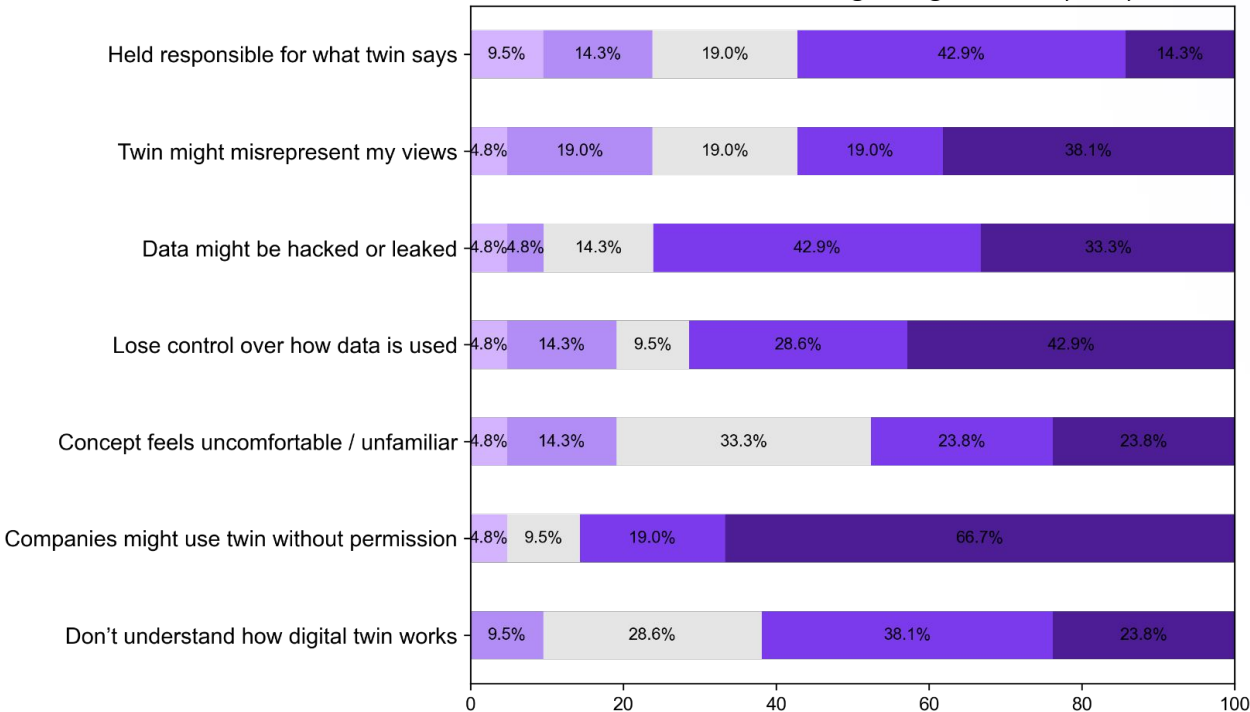


Lowest Comfort Areas:

- **Answer personal opinions:** Only 28.6% comfortable with digital twins expressing personal views/feelings, while 57.2% actively disagree. This represents the highest resistance point.
- **Longer survey to build twin:** 33.4% willing to invest upfront time, but 38.1% resistant, indicating a patience threshold for onboarding.

Quantitative Research:

User Concerns for Digital-Twin Adoption for Research Participation



Response

- Not at all concerned
- Slightly concerned
- Moderately concerned
- Quite concerned
- Very concerned

Most intense concerns (highest “Quite + Very concerned”)

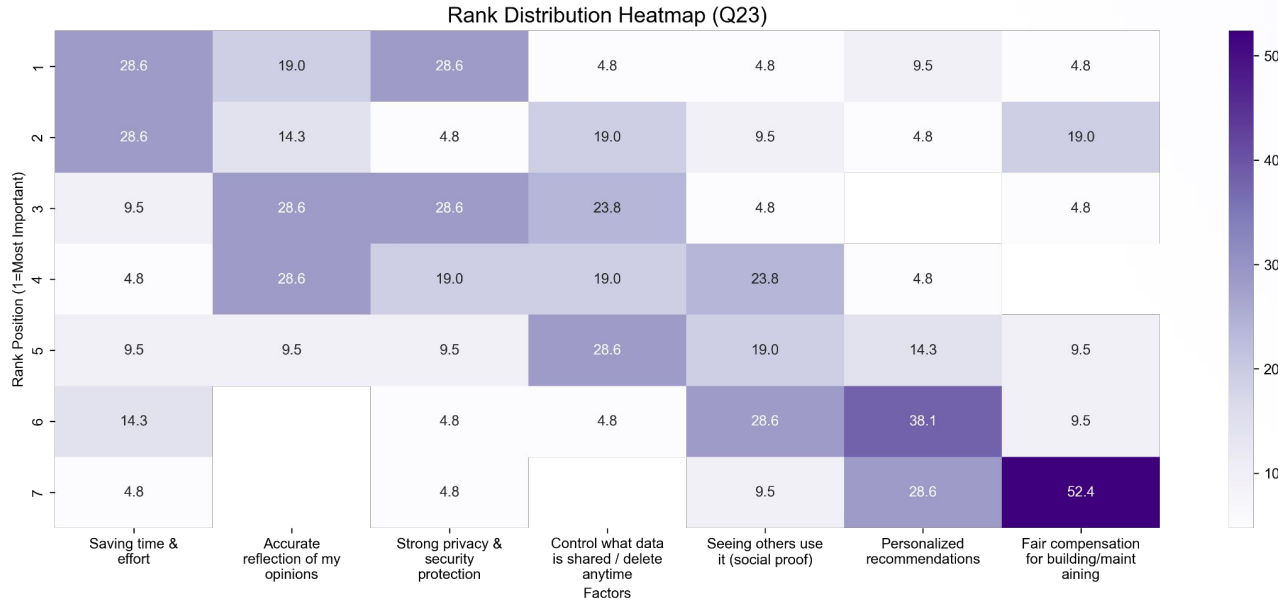
- 1. Unauthorized twin usage** by companies (largest concern cluster).
- 2. Data might be hacked or leaked**, reflecting heightened sensitivity to cybersecurity issues.
- 3. Losing control** over how personal data is used.
- 4. Don't understand how digital twin works**

Strategic Takeaway:

Trust is the primary barrier to adoption. Lower-socioeconomic users are most concerned about unauthorized data use, loss of control, and the overall unfamiliarity of the digital-twin concept itself. Building a clear trust promise and designing a transparent “trust loop” are essential to attract and retain this segment.

Quantitative Research:

User Motivation for Digital-Twin Adoption for Research Participation



- **Fair compensation and personalized recommendations** rank lowest in importance, with 52.4% and 38.1% of users ranking them as least critical (rank 7 and 6 respectively).
- This demonstrates that users **prioritize functional trust** (accuracy, data control, privacy protection) **over external incentives**, whether financial or social."
- **Motivation Paradox:** Financial compensation is the #1 motivator for research participation, yet it is the lowest driver for digital-twin adoption for research participation

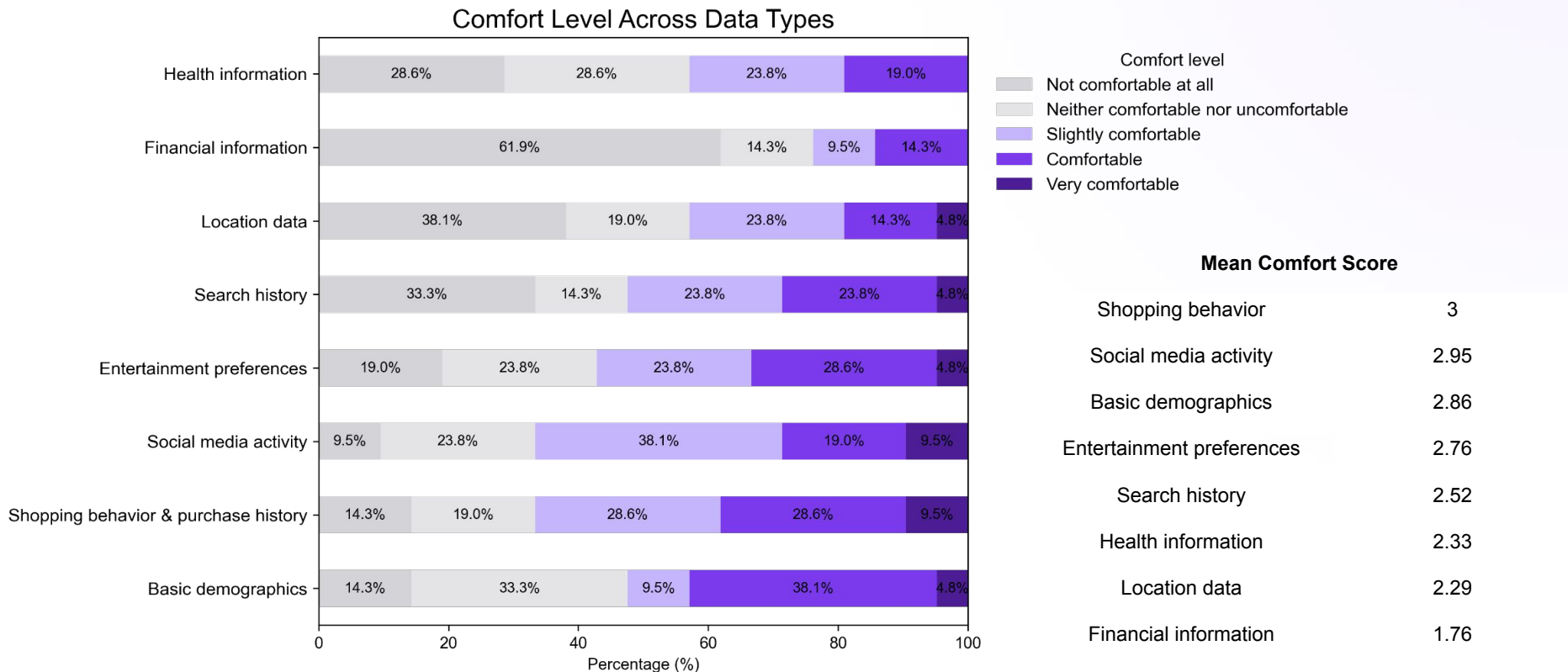
- **Saving time & effort** is the most frequently prioritized as a top-2 factor (chosen by 12 respondents), suggesting it is a primary motivator when deciding whether to use a digital twin.
- However, when considering overall weighted importance (mean rank), **"Accurate reflection of my opinions"** emerges as the highest-ranked factor, indicating that users value the twin's ability to represent them authentically.

Strategic Takeaway:

Digital-twin adoption is not a financial problem—it is a trust problem. Incentives can drive participation, but they cannot overcome trust barriers to adopting the digital twin itself.

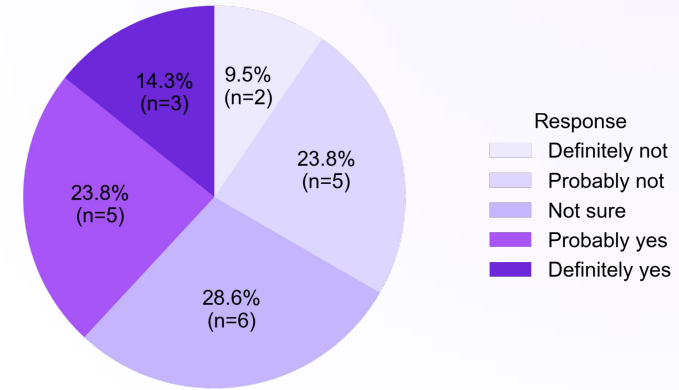
Quantitative Research:

Data Sensitivity Hierarchy: What Users Will (and Won't) Share



Quantitative Research: Key Insights

- **38.1% would likely try it, 33.3% are unlikely, and 28.6% remain undecided.** This indicates a sizable opportunity to convert hesitant users. The undecided segment represents the highest-potential target group for onboarding and education.
- It is noteworthy that Financial compensation **is the #1 motivator** for research participation, **yet it is the lowest driver** for digital-twin adoption for research participation
- Digital-twin adoption is **not a financial problem—it is a trust problem.**
Incentives can drive participation, but they cannot overcome trust barriers to adopting the digital twin itself.
- Lower-socioeconomic users are most concerned about **unauthorized data use, loss of control, and the overall unfamiliarity of the digital-twin concept itself.** Building a clear trust promise and designing a transparent “trust loop” are essential to attract and retain this segment.
- **Overall, adoption likelihood is promising to increase when the platform provides tangible financial value and transparent, user-centric data governance that actively builds trust.**



“
compensates + free , which could earn me money
I don't want another digital me alive
When digital twin mature enough
I would have to see how the process is done
Compensation and guarantees my privacy
Privacy issue about my data
Full control of my data that is shared
”

—When asked what will motivate you to use digital twin

Customer Journey Map

Stage	Mindset	Behaviors / Touchpoints	Pain Points	Opportunities
Awareness (PMOT)	"I see data/tech online but don't know what it means for me."	Users passively encounter the idea on social media or through friends.	The concept feels abstract and 72% worry about losing control of data.	Use simple analogies and start with low-risk data examples users already feel comfortable with.
Interest (ZMOT)	"I'd check it out if it saves time or earns something small."	Users watch short explainers and look for credibility or rewards.	They fear scams and 76% worry a lot about data leaks.	Highlight small rewards and recognizable brands since nearly half would share more data if incentivized.
Evaluation (Deep ZMOT)	"I want to know who's behind this and how my info is used."	Users skim reviews and react strongly to clear or unclear privacy signals.	Their top concern is unauthorized usage and they need accuracy.	Use plain-language privacy, clear opt-ins, and emphasize the "review before submit" control supported by 57% of users.
Trial / Setup (FMOT)	"I'll try it only if it's quick and easy on my phone."	Users start onboarding but drop off when forms feel long or sensitive.	Slow steps or early high-risk data requests cause abandonment.	Keep onboarding under one minute, add tiered consent, and give a small instant reward.
Adoption (SMOT)	"I'll keep using it if it feels simple and worth my time."	Users complete occasional tasks but forget the platform without visible value.	They disengage when benefits aren't obvious or feel one-sided.	Show time saved (a top motivator) and send simple nudges to keep value visible.
Retention (TMOT)	"If it's fair and helpful, I'll tell my friends."	Users share it casually only when the experience stays smooth and trustworthy.	Trust drops quickly if communication stops or anything feels unclear.	Maintain consistent updates, offer referral rewards, and show how user data makes real impact.

Strategic Recommendations: Building Trust + Value

“Building Trust”

1. Make Trust Hyper-Visible and Easy to Understand

Insight: Top fears: unauthorized use, leaks, misrepresentation.

- Use “**Trust Moments**” during onboarding, e.g., “Your data stays private,” “You review before sending” (57% support)
- Replace long privacy text with **5-second explainers** using icons.

2. Close the Loop With Real Outcomes

Insight: If users see nothing happen after giving data → trust collapses.

- Send **monthly impact summaries** (“Your twin answered X surveys; you earned \$Y this month”).
- Show **micro-impact** in the UI so users can clearly see when their twin worked—and how it translated into **payouts**.

3. Partner With Local Anchors to Build Legitimacy

*Insight: Low-SES users trust **community orgs** more than brands.*

- Partner with libraries, unions, and community centers to co-host onboarding; Use **co-branded materials** (“Powered by ____ & Doppel Market”) to boost legitimacy.

“Building Value”

4. Lead With Time Savings + Immediate Utility

Insight: Low-SES users value saving time more than any other practical benefit (top-2 factor for 12 users).

- Position the digital twin as a **low-effort shortcut** that reduces repetitive answering, making research participation easier.
- Messaging: “*Answer once, and your twin handles the rest.*”

5. Provide Instant Rewards + Long-Term User Value

Insight: 47.6% would share more data if rewards are provided.

- Provide instant **\$5–\$8 micro-rewards** for key actions.
- Explore ways the digital twin can **return value to users**, such as more accurate recommendations, personalized insights, or smarter task automation.

6. Add Referral Rewards

Insight: Word-of-mouth is the strongest validation among low-SES users.

- Offer **\$2–\$5 referral credits**.
- Provide **simple sharing tools** (QR code, WhatsApp, SMS).
- Highlight: “*Friends can earn too.*”

Ethical Considerations for Long-Term Success

1. Data Ownership & Informed Consent

- Users must clearly understand what data is collected, when the twin acts, and how it's reused.
- **Low-SES users are especially vulnerable to unclear consent, so Doppel must ensure control is simple, transparent, and genuinely informed.**

2. Representation & Human Dignity

- A twin may still misstate opinions or speak in ways the user disagrees with, especially on sensitive topics.
- To avoid harm or “test subject” feelings, we must **let users review, correct, and control how the twin represents them.**

3. Fairness & Algorithmic Bias

- Biased or incomplete data can misinterpret low-SES behaviors or reinforce stereotypes.
- Doppel must actively audit models to **ensure outputs are fair and do not reproduce existing inequalities.**

4. Accountability & Responsibility

- When the twin produces harmful or incorrect outputs, users need clear recourse and explanations.
- Doppel must own the consequences of its technology and **maintain strong correction and oversight systems.**

Project Limitations

Bias Examples	Qualitative	Quantitative
Unaddressable	Hypothetical Bias (Respondent Bias): "If you were to create something like this, what would you want to know first?" , "What would motivate you to build your digital twin?" This creates the the "Say-Do" Gap. Social Desirability (Respondent Bias): downplayed their desire for money or online engagement to appear more altruistic or sophisticated: <i>Ashley</i>: filled out surveys to "support" people; <i>Carl</i>: an "old school" and not big on social media to distance himself from online drama.	Hypothetical Bias (Response Bias) - Privacy Paradox: Respondents overstate willingness to adopt invasive tech due to the lack of real-world consequences (hypothetical nature). 71% Concerned about data privacy (Q10) yet 67% willing/open to using a Digital Twin that harvests personal data (Q24).
Addressable	Question Sequencing (Survey Design): Across almost all interviewers, the opening question anchors the respondents to specific categories: "streaming, shopping, social media, or using apps", rather than allowing them to recall their organic online behavior.	Demographic Skew (Sampling Bias - Convenience Sampling): The sample over-indexed on a specific sub-demographics (Asian). 52% Asian respondents (vs. ~7.3% National Avg). The sample reflects a bit "Ethnic Bubble" rather than the US market.

Appendix & Supporting Materials

Documentation

Please access our team Google Drive folder here. This contains our research design, raw data, code, and visualizations:

<https://drive.google.com/drive/folders/155cTOVy5mO4xv2ils8jAzyMen9ybnhGI?usp=sharing>